

Unit 3.6 Derivatives of

Note Title

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Logarithmic functions

Section 3.6 p218-223 in Stewart.

Theorem

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

Proof : $y = \ln x$

$$\Rightarrow e^y = x$$

$$\frac{d}{dx} e^y = \frac{d}{dx} x$$

$$\Rightarrow e^y \cdot \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$$

Theorem $\frac{d}{dx} \ln |x| = \frac{1}{x}$

Proof: If $x > 0$ then

$$\frac{d}{dx} \ln |x| = \frac{d}{dx} \ln x = \frac{1}{x}$$

If $x < 0$ then

$$\frac{d}{dx} \ln |x| = \frac{d}{dx} \ln (-x)$$

$$= \frac{1}{-x} \cdot (-1)$$

$$= \frac{1}{x}$$

Theorem

$$\frac{d}{dx} \log_a x = \frac{1}{\log_a x}$$

Proof Self Study.

Example 1

$$\frac{d}{dx} \ln(\sin(x^3))$$

$$= \frac{1}{\sin(x^3)} \cdot \cos(x^3) \cdot 3x^2$$

Example 2

$$\frac{d}{dx} \sin(\ln(1-3x))$$

$$= \cos(\ln(1-3x)) \cdot \frac{1}{1-3x} (-3)$$

Example 3

$$\frac{d}{dt} t^4 \ln(1 - e^{4t})$$

$$= 4t^3 \ln(1 - e^{4t}) + t^4 \frac{1}{1 - e^{4t}} (-e^{4t} \cdot 4)$$

Example 4

$$\frac{d}{dx} \ln(\ln \cos x)$$

$$= \frac{1}{\ln \cos x} \cdot \frac{1}{\cos x} \cdot (-\sin x)$$

Example 5

$$\frac{d}{dx} \ln(\arcsin e^{3x})$$

$$= \frac{1}{\arcsin e^{3x}} \cdot \frac{1}{\sqrt{1 - e^{6x}}} \cdot e^{3x} \cdot 3$$

$$\left\{ \begin{array}{l} \frac{d}{dx} \ln x = \frac{1}{x} \\ \frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx} e^x = e^x \end{array} \right.$$

Example 6

$$\left\{ \frac{d}{dx} \arctan x = \frac{1}{1+x^2} \right.$$

$$\frac{d}{dx} \arctan (\ln(1+4^x))$$

$$= \frac{1}{1+(\ln(1+4^x))^2} \cdot \frac{1}{1+4^x} \cdot \ln 4 \cdot 4^x$$

$$\textcircled{1} \frac{d}{dx} [f(x)]^n = n [f(x)]^{n-1} \cdot f'(x)$$

$$\textcircled{2} \frac{d}{dx} [a^{f(x)}] = \ln a \cdot a^{f(x)} \cdot f'(x)$$

$$\textcircled{3} \frac{d}{dx} [f(x)]^{g(x)}$$

Steps in Logarithmic differentiation

- ① Take natural logs on both sides of $y = f(x)$ & use log laws to simplify
- ② Differentiate implicitly with respect to x
- ③ Solve the equation for y'

Example

$$\frac{d}{dx} [1 - \tan 4x]^5$$

$$= 5(1 - \tan 4x)^4 \cdot (-\sec^2 4x \cdot 4)$$

Example

$$\frac{d}{dx} 4^{\sin 3x}$$

$$= \ln 4 \cdot 4^{\sin 3x} \cos 3x \cdot 3$$

Example

$$\frac{d}{dx} x^x$$

$$\text{Let } y = x^x$$

$$\Rightarrow \ln y = \ln x^x$$
$$= x \ln x$$

$$\ln a^b = b \ln a$$

$$\frac{d}{dx} \ln y = \frac{d}{dx} x \ln x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = 1 \cdot \ln x + x \cdot \frac{1}{x}$$

$$\frac{dy}{dx} = y [\ln x + 1]$$

$$= x^x [\ln x + 1]$$

Example

$$\frac{d}{dx} 4x^{\cos 3x}$$

$$\text{let } y = 4x^{\cos 3x}$$

$$\Rightarrow \ln y = \cos 3x \ln 4x$$

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = -\sin 3x \cdot 3 \cdot \ln 4x + \frac{\cos 3x}{4x} \cdot 4$$

$$\Rightarrow \frac{dy}{dx} = y \left[-\sin 3x \cdot 3 \ln 4x + \frac{\cos 3x \cdot 4}{4x} \right]$$

$$= 4x^{\cos 3x} \left[-\sin 3x \cdot 3 \ln 4x + \frac{\cos 3x \cdot 4}{4x} \right]$$

Example

$$\frac{d}{dx} (\sin 2x)^{x^2}$$

$$\text{Let } y = \sin 2x^{x^2}$$

$$\Rightarrow \ln y = \ln \sin 2x^{x^2}$$
$$= x^2 \ln \sin 2x$$

$$\frac{d}{dx} [\ln y] = \frac{d}{dx} [x^2 \ln \sin 2x]$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = 2x \cdot \ln \sin 2x + x^2 \cdot \frac{1}{\sin 2x} \cos 2x \cdot 2$$

$$\frac{dy}{dx} = y \left[2x \ln \sin 2x + \frac{x^2 \cos 2x \cdot 2}{\sin 2x} \right]$$

$$\frac{d}{dx} (\ln x)^{\cos x}$$

$$y = \ln x^{\cos x}$$

$$\Rightarrow \ln y = \ln \ln x^{\cos x}$$
$$= \cos x \ln \ln x$$

$$\frac{1}{y} y' = -\sin x \cdot \ln(\ln x) + \cos x \cdot \frac{1}{\ln x} \cdot \frac{1}{x}$$

$$y' = \ln x^{\cos x} \left[-\sin x \ln(\ln x) + \frac{\cos x}{x \ln x} \right]$$

Example

$$\frac{d}{dx} \left(\frac{6^x \sin 2x}{e^{9x} \tan 4x} \right)$$

Let

$$y = \frac{6^x \sin 2x}{e^{9x} \tan 4x}$$

$$\ln y = \ln \left(\frac{6^x \sin 2x}{e^{9x} \tan 4x} \right)$$

$$= \ln(6^x \sin 2x) - \ln(e^{9x} \tan 4x)$$

$$= \ln 6^x + \ln \sin 2x - \ln e^{9x} - \ln \tan 4x$$

$$= x \ln 6 + \ln \sin 2x - 9x - \ln \tan 4x$$

$$\ln\left(\frac{a}{b}\right) = \ln a - \ln b$$

$$\ln(a \cdot b) = \ln a + \ln b$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \ln 6 + \frac{\cos 2x \cdot 2}{\sin 2x} - 9 - \frac{\sec^2 4x \cdot 4}{\tan 4x}$$

$$\frac{dy}{dx} = y \left[\ln 6 + \frac{\cos 2x \cdot 2}{\sin 2x} - 9 - \frac{\sec^2 4x \cdot 4}{\tan 4x} \right]$$

Unit 3.7 Self Study